

✓ Global Life Expectancy Notebook

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The purpose of this notebook is to determine the relationship between life expectancy and health expenditure by the government. This research can be helpful in determining if allocating greater government funds to healthcare can improve citizens' outcomes by increasing their life expectancy.

Data is from a Kaggle dataset using data determined by the World Bank:

<https://www.kaggle.com/datasets/mjshri23/life-expectancy-and-socio-economic-world-bank>

Question: Is there a correlation between a country's health expenditure (%) and life expectancy?

Hypothesis: There is no correlation between health expenditure (%) and life expectancy.

- Ho: no correlation between health expenditure and life expectancy
- Ha: there's a positive correlation between health expenditure and life expectancy
- p-value limitation of 0.05

Definitions:

- Health expenditure (%): percent of a country's GDP spent on health expenditures, including healthcare goods and services consumed each year
- Life expectancy: the average life expectancy by the World Bank

1. Import the necessary libraries: numpy, pandas, matplotlib.pyplot, and seaborn

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

2. Read in the data into a file called raw_data

```
raw_data = pd.read_csv('/life_expectancy.csv')
```

raw_data

	Country Name	Country Code	Region	IncomeGroup	Year	Life Expectancy World Bank	Prevalence of Undernourishment
0	Afghanistan	AFG	South Asia	Low income	2001	56.308	
1	Angola	AGO	Sub-Saharan Africa	Lower middle income	2001	47.059	
2	Albania	ALB	Europe & Central Asia	Upper middle income	2001	74.288	
3	Andorra	AND	Europe & Central Asia	High income	2001	NaN	
4	United Arab Emirates	ARE	Middle East & North Africa	High income	2001	74.544	
...	...	...	...	...	...	...	
3301	Vanuatu	VUT	East Asia & Pacific	Lower middle income	2019	70.474	
3302	Samoa	WSM	East Asia & Pacific	Lower middle income	2019	73.321	
3303	South Africa	ZAF	Sub-Saharan Africa	Upper middle income	2019	64.131	
3304	Zambia	ZMB	Sub-Saharan Africa	Low income	2019	63.886	
3305	Zimbabwe	ZWE	Sub-Saharan Africa	Lower middle income	2019	61.490	

3306 rows x 16 columns

3. Clean up the data and create a new dataframe called `clean_data` that only includes relevant information (country name, health expenditure, and life expectancy)

```
clean_data = raw_data.drop(columns=['Country Code', 'CO2', 'Corruption',
clean_data = clean_data.sort_values('Country Name', ascending=True)
clean_data = clean_data[['Health Expenditure %', 'Life Expectancy World
```

clean_data

	Health Expenditure %	Life Expectancy World Bank
2610	11.818562	63.763
1566	8.569672	61.028
1740	8.561907	61.553
2262	9.528871	62.966
1044	9.904675	59.375
...	...	...
2957	7.469752	60.812
1913	8.081738	52.896
2609	7.452066	59.534
3131	8.680062	61.195
3305	7.703565	61.490

2976 rows × 2 columns

- Now, create a pivot table that shows just the health expenditure vs life expectancy (the two variables we are concerned with)

```
clean_data.pivot_table(index='Health Expenditure %', values='Life Expect
```

Life Expectancy World Bank

Health Expenditure %

1.263576	56.288000
1.499940	54.449000
1.518554	54.638000
1.553498	55.945000
1.571148	54.842000
...	...
16.767063	78.787805
16.805836	78.539024
16.844324	78.539024
19.727417	52.372000
20.413414	52.941000

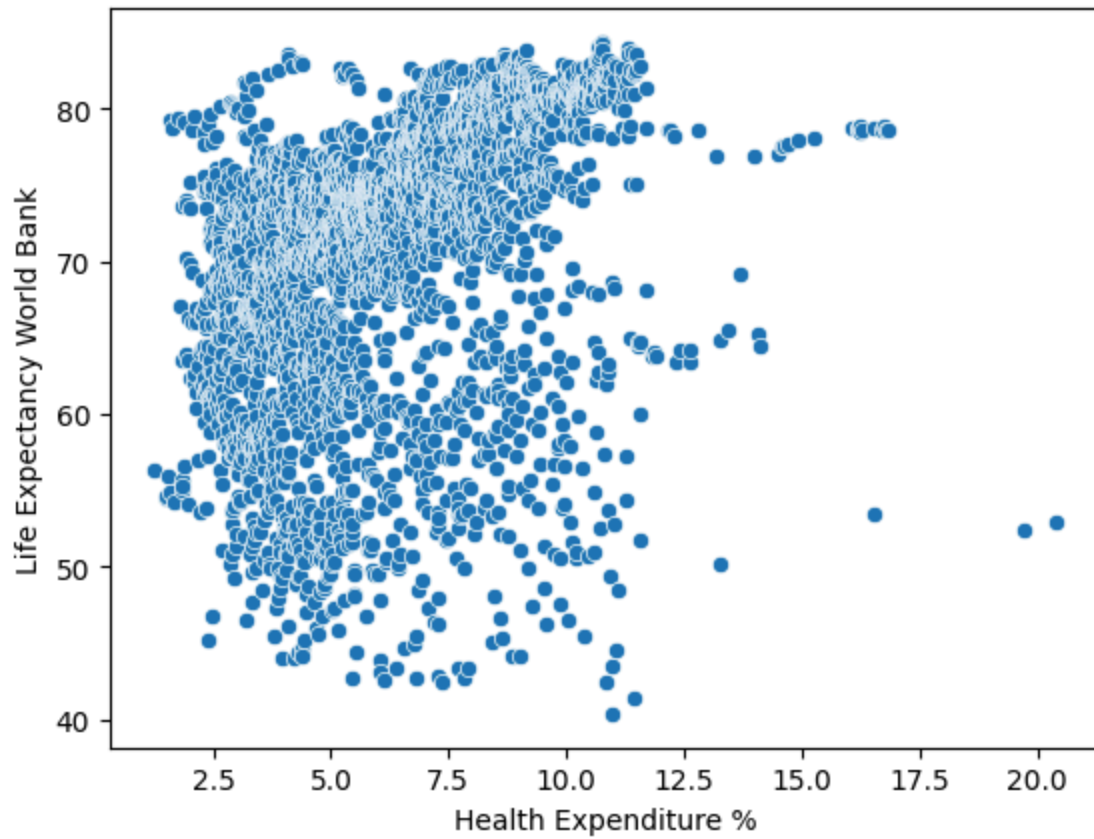
2976 rows × 1 columns

5. Visualize the pivot table into infographics

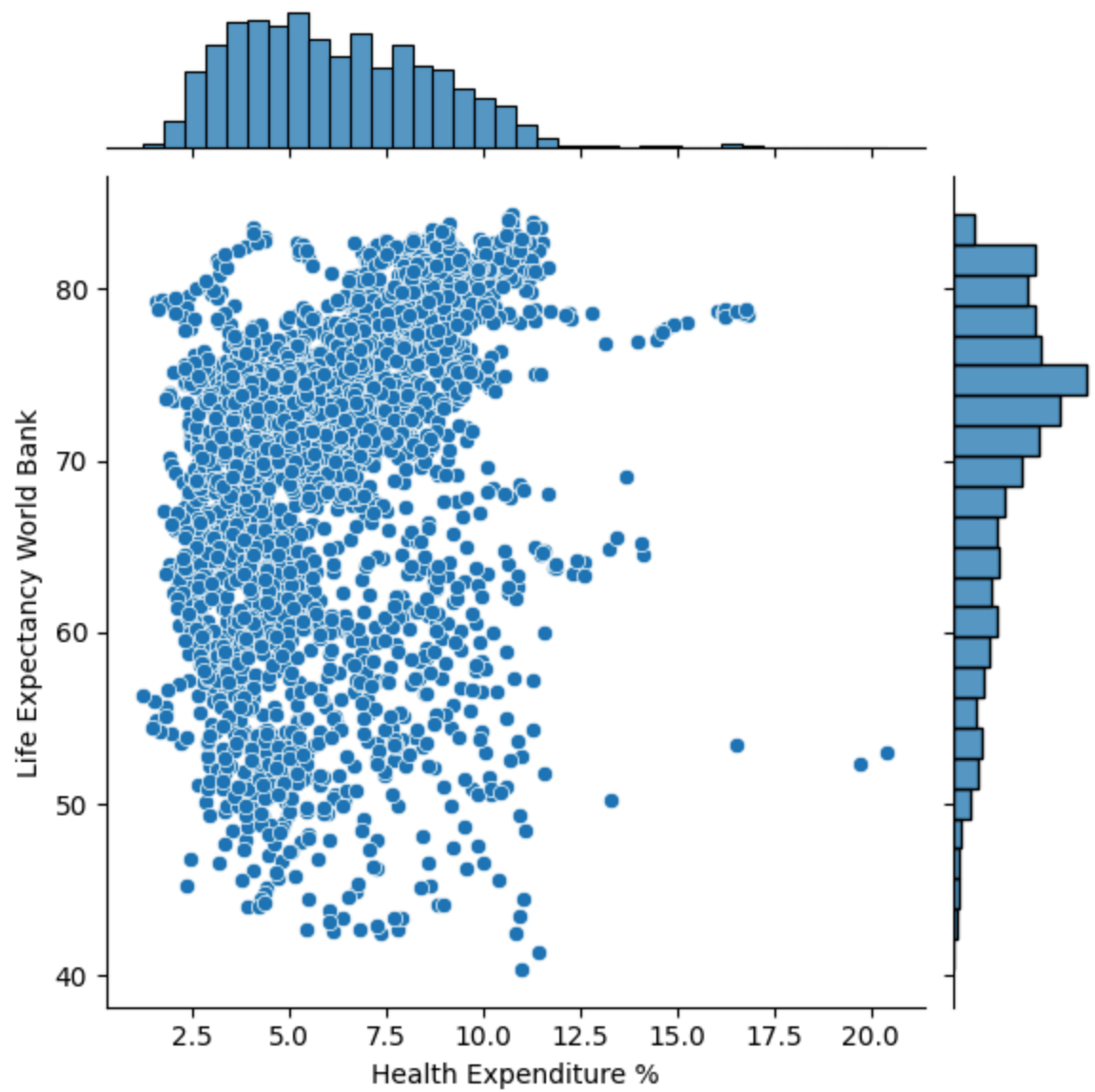
```
comp = clean_data.pivot_table(index='Health Expenditure %', values='Life
```

```
# Scatterplot  
sns.scatterplot(data = comp, x='Health Expenditure %', y='Life Expectanc
```

```
<Axes: xlabel='Health Expenditure %', ylabel='Life Expectancy World Bank'>
```

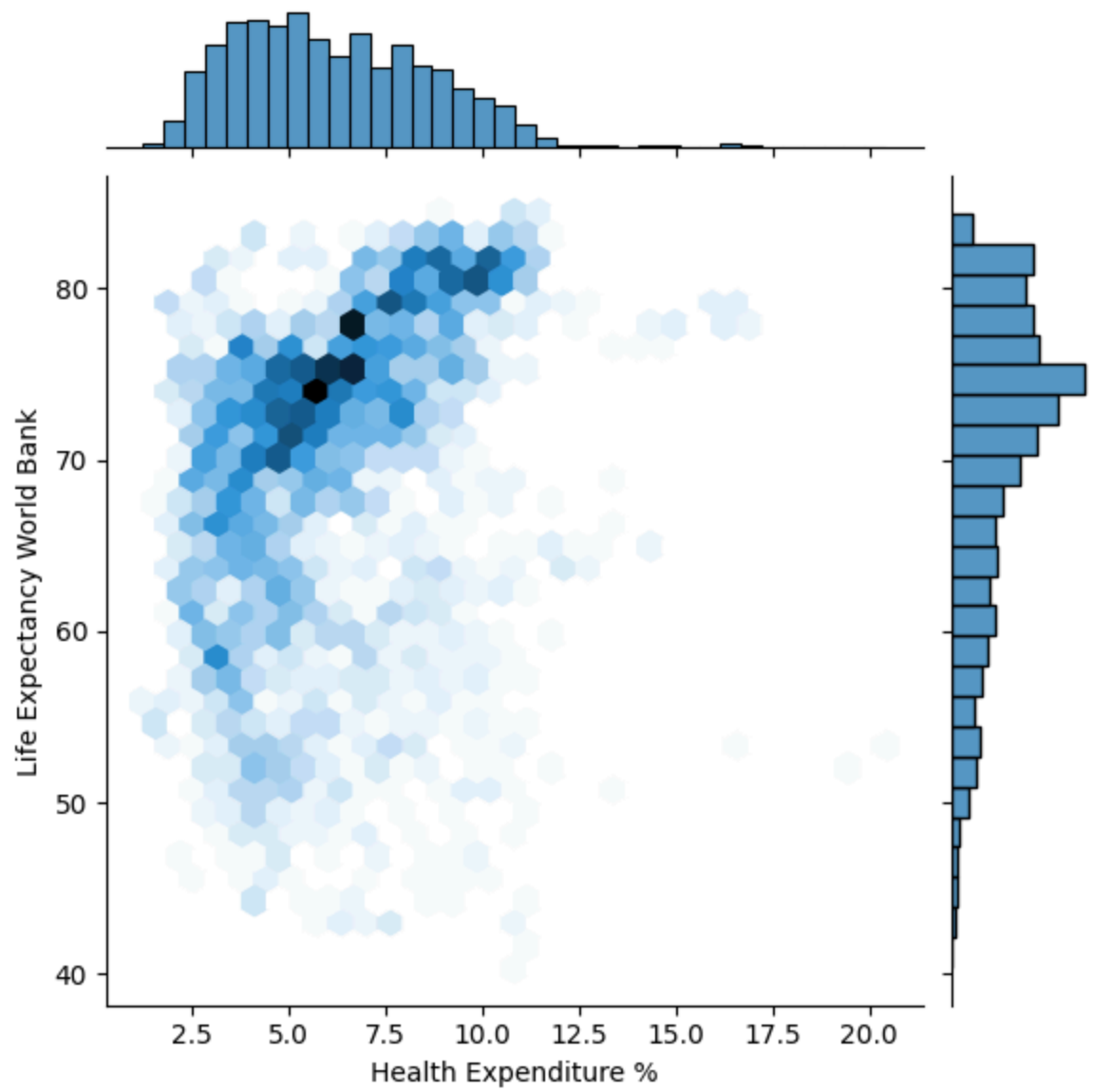


```
# JointPlot  
sns.jointplot(data=clean_data, x='Health Expenditure %', y='Life Expecta
```



```
# Hexbin Plot
```

```
sns.jointplot(data=clean_data, x='Health Expenditure %', y='Life Expecta
```



```
# Kernel Density Estimate Plot
```

```
sns.jointplot(data=clean_data, x='Health Expenditure %', y='Life Expecta
```

